

1917, German Gothas heavy bombers attacked London and Paris, and, by the war's end, the UK and France were mounting their own bombing raids against German cities. After the war, some air commanders argued – wrongly – that future wars would be won by strategic bombing alone.

NEW AIRCRAFT

World War I airmen fought mostly in wood-and-canvas biplanes. During the 1930s, air forces were transformed by streamlined monoplanes – many of all-metal construction – with more powerful engines and improved performance. These new aircraft, including the Messerschmitt Bf 109 fighter and the dive-bombing Junkers Ju 87 Stuka, were first used by German forces in the Spanish Civil War (1936–39).

In the early years of World War II, from 1939 to 1941, Germany achieved a series of victories through coordinated air–ground warfare. This was based on radio contact between tanks and aircraft, operating as “aerial artillery”. They also pioneered the use of airborne troops, landed by parachute or glider. However, the UK's radar-based air defence system enabled the Royal Air Force to prevent the *Luftwaffe* gaining command of the air in the Battle of

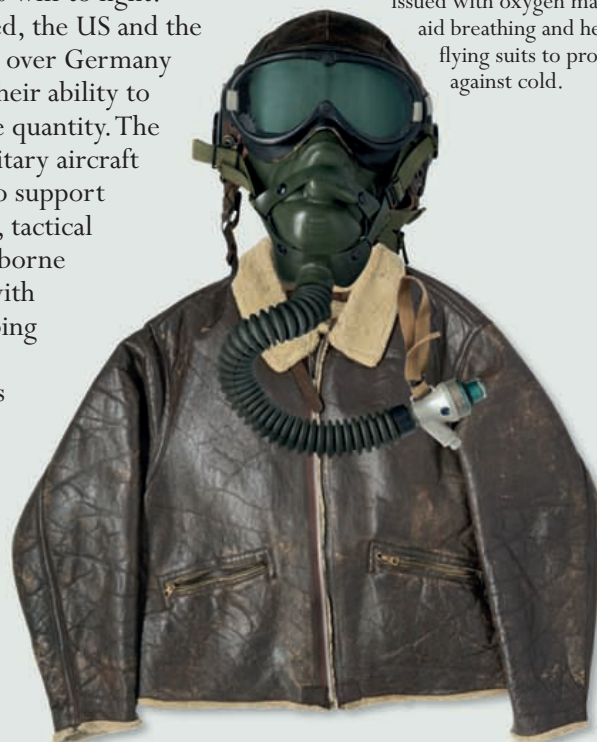
Britain, in the summer of 1940. A sustained night bombing campaign against British cities, from September 1940 to May 1941, caused devastation, but failed to break the UK's will to fight.

As World War II progressed, the US and the UK achieved air superiority over Germany and Japan, chiefly through their ability to manufacture aircraft in huge quantity. The US built about 300,000 military aircraft during World War II, used to support operations in ground attack, tactical bombing, transport, and airborne troop landing roles, along with independent strategic bombing offensives. A day-and-night campaign by Allied bombers eventually reduced German cities to rubble, although still not to decisive effect.

Innovations such as the first jet aircraft or the first military helicopters were of marginal effect, but the arrival of the atomic bomb in 1945 heralded a new era in warfare.

▼ BOMBER CREW KIT

In World War II, American B-17 bomber crews flew at high altitude in unpressurized aircraft. They were issued with oxygen masks to aid breathing and heated flying suits to protect against cold.



KEY FIGURE

BARON VON RICHTHOFEN
1892–1918

Germany's most famous World War I air ace, Manfred von Richthofen served as cavalry officer before becoming a fighter pilot. Credited with downing 80 enemy aircraft – more than any other pilot in the war – the “Red Baron” was himself shot down and killed in April 1918.



▲ Von Richthofen led a squadron that was nicknamed the “Flying Circus” due to its planes' bright colours; the baron himself flew a red Fokker Dr.I triplane.

◀ AMERICAN BOMBERS

A medium-range Mitchell B-25 bomber flies a bombing raid in France during World War II. Air superiority proved essential to the success of armies in combat.

